INFOTACT SOLUTIONS|UI/UX INTERNSHIP PROJECT

OmniSpace

Spatial UI System

Week 3 Report: Z-axis Logic

Depth, Distance and Scale in Spatial Interfaces

UX Research and Spatial Architecture Report



Conceptual illustration of depth layers

PROJECT
OmniSpace – Spatial UI System

WEEKANDTOPIC
Week 3 – Z-axis Logic

MYROLE
UX Research & Spatial Architecture

TEAMPOSITION
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PROGRAMME
Infotact Solutions – UI/UX Internship

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About This Report

ThisreportpresentstheWeek3researchon**Z-axislogic**fortheOmniSpacespatialUIsystem.Itwas prepared by Member 1, whose responsibility is UX research and spatial architecture. It is based on general UX principles, basic geometry and recommended reading. It does **not** contain user-testing results, device measurements or a completed implementation.

To keep the evidence clear, the report uses the following labels:

> **DesignRecommendation**
> A proposal made by the author, based on general UX and ergonomic principles. It is a starting point for the team and still needs to be validated through prototyping and testing.

> **VerificationNote**
> A point where exact values (for example distances, sizes or angles) depend on the target device or platform. These values are not stated here and must be taken from the official platform guidelines.

> **ConceptualIllustration**
> Every diagram in this report is an original, schematic drawing made to explain an idea. Diagrams are **not** to scale and do not show real device output or measured data.

Important:thisreportdoesnotclaimanyverifiedplatformspecification.Theonlynumbersusedare(a) a web contrast standard cited as a baseline in Section 5 and (b) a worked geometry example with hypothetical values in Section 4.

1. Introduction to Z-axis Logic in Spatial UI

Spatial interfaces, such as those used in augmented reality (AR), mixed reality (MR) and virtual reality (VR), place user interface elements inside the three-dimensional space around the user. A normal screenhastwoaxes:**X**(lefttoright)and**Y**(toptobottom).Aspatialinterfaceaddsathirdaxis,**Z**,which describes **depth**: how near or far an element appears from the user's eyes.

Z-axis logic is the set of design rules that decide:

- how far from the user each UI element should be placed;
- how large the element should be at that distance; and
- which elements appear in front of or behind other elements.

Project context

OmniSpaceisthespatialUIsystembeingdesignedinthisinternshipproject.TheWeek3topicisZ-axis logic. As Member 1, my role is **UX research and spatial architecture**: studying how depth should be organised so that the interface is readable, comfortable and easy to understand. The findings are intended to support the wider design work on the OmniSpace system.

Why the Z-axis matters

- **Hierarchy:**in2D,importanceisshownwithsize,colourandposition.In3D,distanceandlayering also communicate importance.
- **Readability:**thesamepanellookssmallerasitmovesaway,sotextandbuttonsthatarereadable up close may become hard to read at a distance.
- **Comfort:**placingcontenttoonearortoofar,orchangingdepthtoooften,cantiretheeyesand neck.
- **Consistency:** clear depth rules help every screen in OmniSpace behave in a predictable way.

Basis and limits of this report

The content is built from desk research and conceptual design thinking. Where a recommendation dependsonaspecificheadsetorplatform,itismarkedasneedingverification.Thereportdescribes *what should be decided and tested*, not results that have already been achieved.

2. Objectives of Z-axis Positioning

TheaimofZ-axispositioningistogiveeveryUIelementasensibledepth,sizeandlayersothatusers can find, read and use it without effort. The main objectives are listed below.

Objective	Whyitmatters	Expecteduseforthteam
1.Defineadepthvocabulary	Everyoneneedsthesamewords for depth positions.	Shared terms: Near, Middle, Far; Foreground,MiddleGround,Background.
2.Assignrolestodistances	Differenttaskssuitdifferent distances.	Asimpleguideforchoosingwhereapanel should sit.
3.Keepcontentreadable	Apparentsizechangeswith distance.	Aruleforscalingtextandbuttonsby viewing distance.
4.Supportclearhierarchy	Usersmustknowwhatisimportant at a glance.	Alayeringmodelforcontrols,contentand supporting information.

Objective	Whyitmatters	Expecteduseforthteam
5.Protectusercomfort	Poordepthchoicescancause fatigue.	Achecklistofergonomiccautionsforlater testing.
6.Createatestableruleset	Designrulesshouldbeconfirmed with real users.	Recommendationsthatcanbeprototyped and validated.

3. Near, Middle and Far UI Panel Positioning

A simple way to organise depth is to group panels into three zones based on their distance from the user:**Near**,**Middle**and**Far**.Eachzoneissuitedtoadifferentkindofcontent.Figure1showsthisidea from the user's point of view.

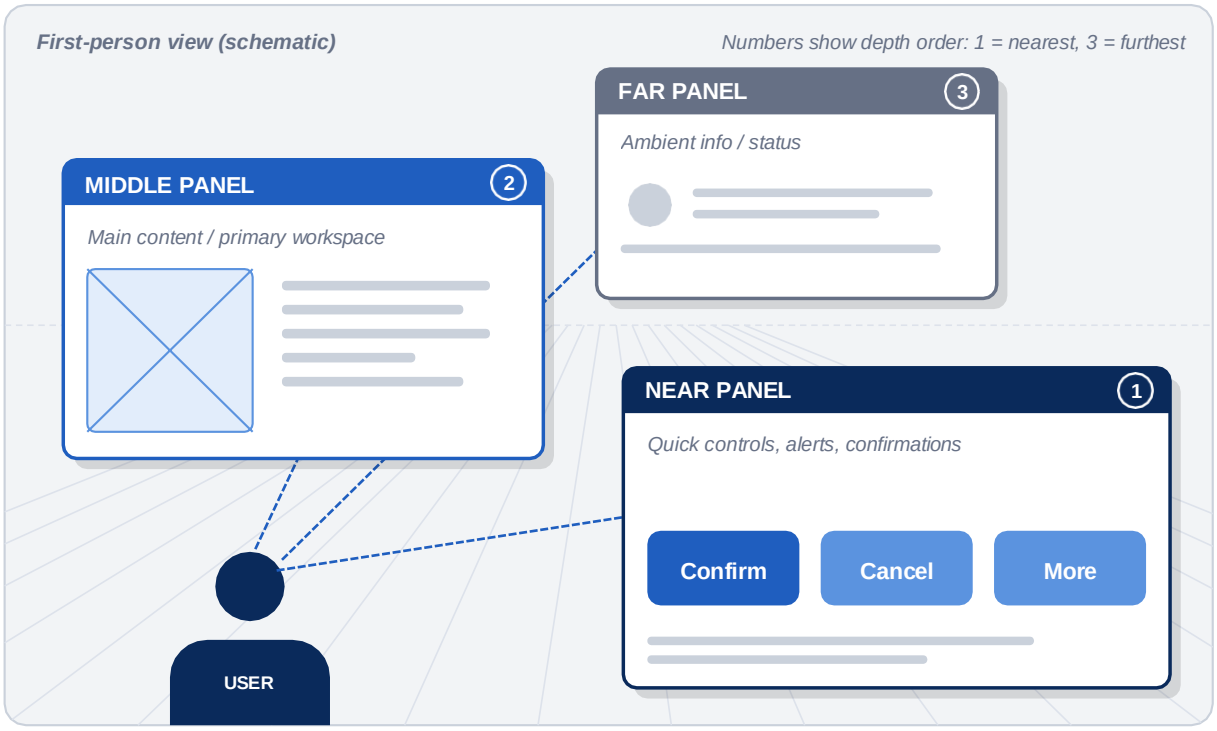


Figure 1. *Near, Middle and Far UI panels floating in front of a user.* ***Conceptual illustration – not to scale.*** *Distances and sizes are schematic.*

Possible purposes of each zone

Thetablebelowdescribesthepossiblepurposeofeachpanel.Thesearedesignproposalsbasedon general UX reasoning.

Panel	Possiblepurpose	Examplecontent	Cautions
NearPanel	Short,quickinteractions that need immediate attention.	Confirmations,alerts,quick toggles, short tool menus.	Avoidlongreading.Contentthat is too close can strain the eyes and block the view.
MiddlePanel	Themainworkingareaand primary content.	Documents,media,theactive task, main navigation.	Shouldbethemostcomfortable zone. Do not overcrowd it.
FarPanel	Information checked occasionally,orcontextfor the scene.	Statusinformation,overview, reference material, environment context.	Textmustbescaledup.Avoid precise or critical interactions here.

Moving content between zones

- Let the user move or pull a panel closer or push it away, instead of moving it automatically.
- Use smooth, short transitions so the user can follow the change in depth.
- Keep the same panel type in the same zone across screens so users build a stable mental model.

> **VerificationNote**
> The actual distance range for “near”, “middle” and “far” differs between devices and platforms. It is deliberately **not** fixed in this report. It should be taken from the official guidelines of the target device and then confirmed in testing.

4. UI Scaling Based on Viewing Distance

Apparent size and visual angle

A user does not judge size by the real width of a panel, but by how large it appears to the eye. This is describedbythe**visualangle**.Foraflatpanelfacingtheviewer,thevisualanglecanbecalculatedas:

Visual angle = 2 × arctan(panel height ÷ (2 × viewing distance))

Whendistanceincreasesandthepanelsizestaysthesame,thevisualanglebecomessmaller.The panel looks smaller, and the text on it becomes harder to read.

Worked geometry example

Theexamplebelowuses**hypothetical**valuestoshowtheeffect.Theyarenotmeasurementsfromany device.

Viewingdistance	Apparentsizeofa0.4mtallpanel	Panelheightneededtokeepthe1m apparent size
1m	about22.6°	0.4m
2m	about11.4°	0.8m
4m	about5.7°	1.6m

*Readingthetable:*doublingthedistanceroughlyhalvestheapparentsizeofapanel.Tokeepthesame apparent size, the panel height must grow in proportion to the distance.

Visual summary

Figure2showsthesameideaasasideview.RowAkeepsthepanelsizefixed,whileRowBincreases panel size with distance.

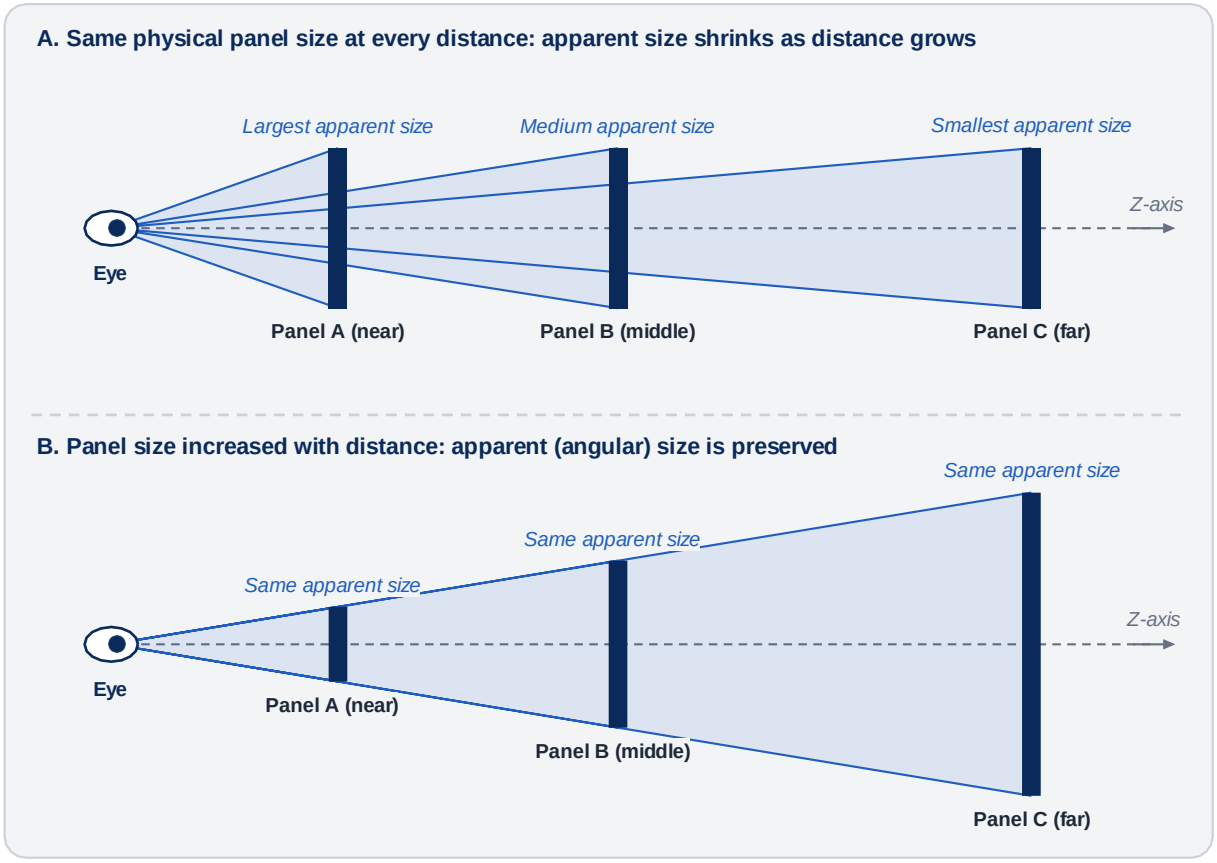


Figure 2. *UI scaling with viewing distance (side view). Row A: equal panel size gives a shrinking apparent size. Row B: panel size grows with distance to keep the same apparent size.* ***Conceptual illustration – not to scale.***

Two scaling approaches

Approach	Advantage	Limitation
Fixedphysicalsize (panelkeepsitsreal-worldsize)	Feelsnatural.Givesastrongsense of depth.	Farpanelsbecomesmallandtext may become unreadable.
Distance-compensatedsize (panelgrowswithdistance)	Keepsapparentsizeand readability stable.	Reducesthenaturaldepthcue.Large far panels may fill too much of the view.

How designers should preserve readability

- Definetheminimumsizeoftextandbuttonsby**visualangle**(apparentsize),notbypixelsor physical size alone.
- Increasepanelandtextsizeforfarpanels,butsetamaximumsothepanelstayswithina comfortable field of view.
- Useahybridapproach:scalepartlywithdistancesothatdepthisstillvisible,butneverbelowthe minimum readable size.
- Keep long reading text in the Middle zone, where it is easier to control size and comfort.

DesignRecommendation
Treat “minimum text angle” and “minimum target angle” as design tokens in OmniSpace. Each panel then calculates its own physical size from its distance. The actual token values should come from platform guidelines and be confirmed by testing.

5. Text Readability and Button Visibility

Textandbuttonsarethepartsoftheinterfacethatusersdependonmost.Depthcanreducetheir clarity, so they need special attention.

Text readability

- **Size:** keep text above a minimum visual angle at its viewing distance (see Section 4).
- **Weightandtypeface:**preferclearsans-seriffontswithregularormediumweight.Avoidverythin strokes, which can shimmer or vanish on a display.
- **Contrast:**usestrongcontrastbetweentextanditsbackground.Placetextonasolidorsemi-solid backing plate instead of directly on a changing background.
- **Length:** use short lines and short paragraphs. Keep long reading in the Middle zone.
- **Placement:**avoidputtingimportanttextintheBackgroundlayeroronFarpanelsthattheuser rarely looks at.

Button visibility

- **Targetsize:**makebuttonslargeenoughtoselectcomfortably,andscalethemwithdistanceinthe same way as text.
- **Spacing:** leave clear gaps between buttons so that a nearby target is not selected by mistake.
- **States:**showclearhoverorfocus,pressed,selectedanddisabledstates.Asmallforward movement or highlight can help show which control is active.
- **No hidden controls:** make sure another panel never covers an active button.
- **Placementbydistance:**placefrequentcontrolsintheForegroundlayerorNearzone,butkeep them out of the centre of the main content.

Summary checklist

Element	Guideline	Reason
Bodytext	Meetstheminimumvisualangleforits distance.	Apparentsizedropsasdistance grows.
Headingsandlabels	Largerandbolderthanbodytext.	Helpsusersscanquickly.
Textbackground	Useabackingplatewithstrongcontrast.	Realorvirtualbackgroundscan change behind a panel.
Buttons	Largetargetswithcleargapsandvisible states.	Reducesselectionerrorsandeffort.
Farcontent	Scaleup,simplify,limittonon-critical information.	Detailandpreciseinputareharderata distance.

> **VerificationNote**
> Asastartingbaselineforcontrast,theW3CWebContentAccessibilityGuidelines(WCAG)askfora contrast ratio of at least 4.5:1 for normal text and 3:1 for large text and interface components. WCAG is a**web** accessibility standard. It is used here as a reference point only, and spatial platforms may publish their own guidance.

6. Depth Layering in Spatial Interfaces

Depth layering means arranging the elements of one interface into layers at different depths, so that the orderofimportanceiseasytosee.Thisreportproposesthreelayers:**Foreground**,**MiddleGround**and **Background**.

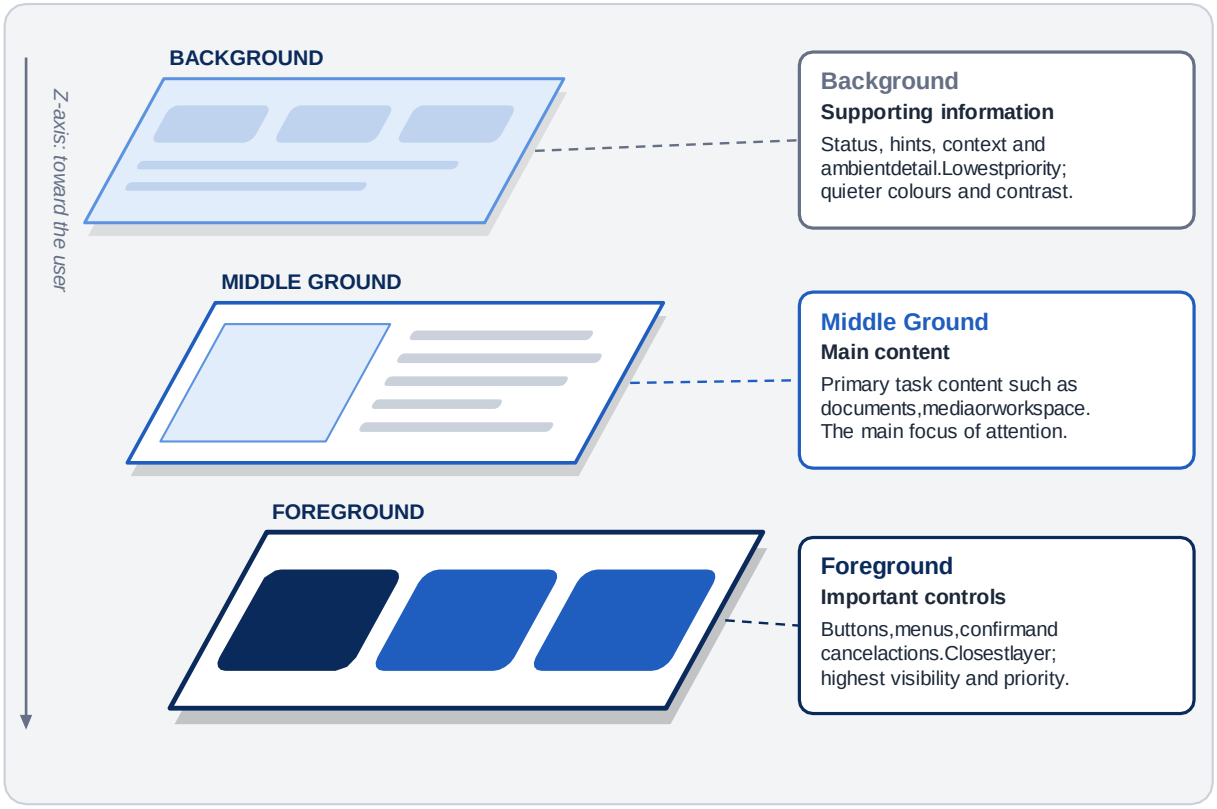


*Figure 3. Depth layering: Foreground (important controls), Middle Ground (main content) and Background (supporting information). **Conceptual illustration – not to scale.***

Role of each layer

Layer	Role	Designnotes
Foreground	**Importantcontrols**:mainactions, menus, confirmations.	Highestcontrastandclarity.Keepsmallsoit does not block the main content.
MiddleGround	**Maincontent**:thematerialtheuseris working with.	Largestandmoststablelayer.Keepitina comfortable viewing position.
Background	**Supportinginformation**:status, hints, context.	Calmercoloursandloweremphasis.Nocritical actions here.

Depth cues to support the layers

Usersunderstanddepthfromseveralcues.Designerscancombinethemsothatthelayeringiseasyto read:

- **Relative size:** nearer items appear larger.
- **Overlap:** an item in front hides part of an item behind it.
- **Shadow and elevation:** a soft shadow shows that a layer floats above another.
- **Transparency or softness:** background layers can be calmer or slightly softened.
- **Motion:** small movement as the user's head moves helps show separation between layers.

Thecuesshouldagreewitheachother.Forexample,alayershouldnotlookbiggerandalsolook further away at the same time.

Zones and layers are different ideas

	Near/Middle/Far(Section3)	Foreground/MiddleGround/Background (Section 6)
Describes	Thedistanceofseparatepanelsfrom the user.	Thepriorityofelementsinsideoneinterface composition.
Mainquestion	Whereshouldthispanelsitintheroom?	Whatshouldbeinfrontfortheuser'sattention?
Usedtogether	A Middle Panel can itself hold a Foreground control strip, a Middle GroundcontentareaandaBackground status bar.	

> **DesignRecommendation**
> Limit each composition to three depth layers. More layers may be harder to understand and may add visual clutter, but this limit should be tested in a prototype.

7. User Comfort and Ergonomic Considerations

Comfortaffectshowlongandhowwellpeoplecanuseaspatialinterface.Z-axisdecisionshaveadirect effect on comfort.

Focusing and eye comfort

Many head-mounted displays show images at a fixed focus distance, while the two eyes turn inward or outwardtomatchtheapparentdepthofanobject.Whenthesetwocuesdonotmatch,thisiscalledthe **vergence–accommodation conflict**. It is a well-documented source of visual fatigue (see Hoffman et al.,2008,andShibataetal.,2011inSection11).Contentplacedveryclosetotheeyesismorelikelyto make this problem noticeable.

Head, neck and body comfort

- Placethemaincontentclosetotheuser'snaturallineofsight,sothatfrequentheadturningand neck bending are not needed.
- Avoid arranging important panels far to the left or right, above or below the user's normal view.
- Avoidrequiringtheusertoholdtheirarmsupforlongperiodstoreachcontrols(oftencalled"arm fatigue"). Keep frequent actions easy to reach.

Motion and change of depth

- Avoid sudden or fast changes of depth. Use gentle transitions the user can follow.
- Avoidconstantmovementorflashingin theBackgroundlayerthatattractsattentionawayfromthe task.
- Giveusersthechoicetorepositionpanels,becausebodysize,postureandvisiondifferfromperson to person.

Accessibility and different users

- Provide options to scale text and panels, and to adjust contrast.
- Donotusedepthalonetocarrymeaning.Supportitwithlabels,iconsorcoloursothatuserswith limited depth perception can still understand the interface.
- Consider seated and standing use, and short and long sessions.

> **VerificationNote**
> Recommended comfortable viewing distances, minimum distances and placement angles are published by device makers and vary by device. They are not set here. They should be checked in the official guidelines listed in Section 11 and confirmed in usability testing.

8. Common Problems and UX Solutions

Thetablelistsproblemsthatarecommonlyexpectedindepth-basedlayouts,togetherwithproposed solutions. These are design considerations, not results from testing.

Problem	Whyithappens	ProposedUXsolution
Contentistooclosetothe user	Apanelisplacedinsidea hard-to-focus distance.	KeepNearcontentshortandquick.Letthe user push it away. Use platform minimum distances.
Texttoosmallonfarpanels	Apparentsizefallswithdistance.	Defineminimumvisualangles.Scalefar panels up within a limit.
Panelsoverlapandhide controls	Depthorderandpositionwerenot planned.	KeepactivecontrolsintheForeground.Do not allow another panel to cover them.
Toomanydepthlayers	Depthusedfordecorationinstead of meaning.	Useatmostthreelayerspercomposition. Removelayersthatdonotcarrymeaning.
Lowcontrastonchanging backgrounds	Real-worldorscenebackgrounds vary.	Usebackingplatesandstrongtextcontrast.
Conflictingdepthcues	Size,shadowandpositionsuggest different depths.	Makethecuesagree.Revieweach composition for consistency.
Neckoreyefatigue	Panelsspreadtoowide,ordepth changes often.	Keepmaincontentnearthelineofsight.Use smooth, infrequent depth changes.
Buttonshardtoselect	Targetsaresmallortooclose together.	Scaletargetsbydistance.Addspacingand clear visible states.
Inconsistentdepthbetween screens	Noshareddepthrules.	Applythesamezoneandlayerrulesacross OmniSpace.

9. UX Design Recommendations

Thefollowingrecommendationsareproposalsmadebytheauthorforthe OmniSpaceteam.Theyare **not verified platform specifications** and should be confirmed through prototyping and testing.

Recommended Z-axis design rules

ID	Recommendation
R1	Usethreedepthzones(Near,Middle,Far)withclearpurposes:quickactions,maincontentandambient information.
R2	PlacetheprimarytaskcontentintheMiddlezone,neartheuser'snaturallineofsight.
R3	KeepNearpanelsshort-livedandsimple.Donotusethemforlongreading.
R4	Definetextandbuttonsizesbyvisualangleandleteachpanelcalculateitsphysicalsizefromits distance.
R5	Useahybridscalingapproachwithaminimumreadablesizeandamaximumpanelsize.
R6	OrganiseeachcompositionintoForeground(controls),MiddleGround(content)andBackground (support).
R7	Makedepthcuesagreewitheachother(size,overlap,shadow,position).
R8	Usebackingplatesandstrongcontrastfortextandcontrols.
R9	Letusersmove,scaleandresetpanels.Avoidautomaticorsuddendepthchanges.
R10	Donotrelyondepthalonetocarrymeaning.Addlabels,iconsorcolourassupport.
R11	Recordalldistance,sizeandanglevaluesinonedesign-tokentableonceconfirmedfromplatform guidelines.

Suggested validation steps (proposed, not yet carried out)

- Checktheofficialguidelinesofthetargetplatformandrecordtheconfirmeddistance,sizeand comfort values.
- Build a simple prototype with Near, Middle and Far panels and three depth layers.
- Testtextreadabilityandbuttonselectionatseveraldistanceswithasmallgroupofusers,using clear tasks.
- Collect comfort feedback after each session and revise the rules.
- Update this report with real findings once testing has been completed.

10. Conclusion

Z-axis logic gives structure to the third dimension of a spatial interface. By grouping panels into Near, Middle and Far zones, scaling content by viewing distance, and organising each composition into Foreground,MiddleGroundandBackgroundlayers,OmniSpacecanpresentinformationinaclearand comfortable way.

ThisWeek3reportprovidesaconceptualfoundationforthatwork.Itexplainsthemainideas,describes the risks to readability and comfort, and proposes design rules. It does not report user-testing results or device measurements. The exact distances, sizes and angles are intentionally left open, because they must be verified against the target platform and tested with real users.

Therecommendednextstepistoconfirmtheplatformvalues,buildasmallprototypeofthethree-zone and three-layer model, and test it. The results can then be used to refine the rules for the OmniSpace system.

11. References and Further Reading

The sources below are recommended reading for platform specifications and for the research behind the comfort and readability topics. Platform documentation changes over time, so the current version shouldbecheckedandanaccessdaterecordedbeforeitiscitedformally.Thisreportdoesnotquote numerical values from these sources.

Platform design guidance (check for exact specifications)

[1] AppleInc.*HumanInterfaceGuidelines:DesigningforvisionOS.* developer.apple.com/design/human-interface-guidelines

[2] MetaPlatforms.*MetaQuestdesignanddeveloperdocumentation.*developers.meta.com

[3] Microsoft. *Mixed Reality design documentation* (comfort, content placement and text guidance). learn.microsoft.com

[4] Google.*AndroidXRdesignanddeveloperguidance.*developer.android.com

Research and standards

[5] Hoffman, D. M., Girshick, A. R., Akeley, K., & Banks, M. S. (2008). Vergence–accommodation conflicts hinder visual performance and cause visual fatigue. *Journal of Vision, 8*(3), 33.

[6] Shibata, T., Kim, J., Hoffman, D. M., & Banks, M. S. (2011). The zone of comfort: Predicting visual discomfort with stereo displays. *Journal of Vision, 11*(8), 11.

[7] Fitts, P. M. (1954). The information capacity of the human motor system in controlling the amplitude of movement. *Journal of Experimental Psychology, 47*(6), 381–391.

[8] Jerald,J.(2015).*TheVRBook:Human-CenteredDesignforVirtualReality.*ACMBooks/Morgan&Claypool.

[9] W3C.*WebContentAccessibilityGuidelines(WCAG)2.2.*w3.org/TR/WCAG22/